Complete Data Analysis Masterclass

Hands-On Mentorship: Python, SQL, Power BI, Excel & Tableau

Instructor: Shubham Singh - Lead Python & Data Architecture Consultant
Industry Background: Ducat, Netmax Technologies, LNP Infotech | 120+ Senior Developers & Students Mentored

BATCH SCHEDULE:	TIMINGS:	FORMAT & DURATION:
Saturday & Sunday Only	6:00 PM - 8:00 PM IST	8-10 Weeks (Live Online Hands-On)

1 Excel Advanced Modeling & Cleansing

- Advanced Lookup & Formulas: XLOOKUP, INDEX/MATCH, VLOOKUP, Nested IFs, SUMIFS
- Data Cleaning, Text Functions (TRIM, SUBSTITUTE, TEXTSPLIT), and Flash Fill
- Interactive Pivot Tables, Calculated Fields, Grouping & Executive Pivot Charts
- Data Validation, Conditional Formatting & What-If Financial Analysis

2 SQL for Data Analysis & Relational Databases

- Relational Database Architecture, Schema Design, Normalization, Primary/Foreign Keys
- Complex Querying: Multi-Table INNER/LEFT/RIGHT/FULL JOINs, Subqueries, and CTEs
- Analytical Window Functions: ROW_NUMBER, RANK, DENSE_RANK, NTILE, LEAD/LAG
- Aggregations, Grouping Sets, HAVING clauses, and Real-world Query Optimization

3 Python Programming for Data Analytics

- Python Foundations: Data structures (Lists, Tuples, Dictionaries), Control Flow, Functions
- NumPy Computing: Vectorized array operations, matrix mathematics, indexing & slicing
- Pandas DataFrames: Reading CSV/Excel/SQL, Filtering, Handling Missing/Duplicate records
- Automated Data Transformation, Aggregations, GroupBy pipelines, and Scripted Reporting

4 Power BI Business Intelligence & Interactive Dashboards

- End-to-End ETL with Power Query: Cleansing, Unpivoting, Merging & Appending Queries
- Relational Data Modeling: Star Schema, Snowflake Schema, 1-to-Many Relationships
- DAX Measures: CALCULATE, FILTER, ALL, RELATED, Date-Time Intelligence (YTD, MTD, YoY)
- Executive Visualization: Drill-through reports, interactive slicers, KPI visual cards

5 Tableau Visual Storytelling & Dashboards

- Connecting to Varied Sources (Files, Relational Databases, Cloud Warehouses)
- Visual Dimensions vs Measures, Discrete vs Continuous data attributes
- Calculated Fields, Table Calculations, User Parameters, and Custom Sets
- Multi-layer Interactive Dashboards, Action Filters, and Executive Storyboards

HANDS-ON PRODUCTION CAPSTONE PROJECTS:

1. Retail & E-Commerce Sales Performance Suite (Python ETL + SQL + Power BI Dashboard)
2. Financial Transactions & Compliance Auditing Analysis (SQL + Tableau Visual Storyboard)

All students complete live portfolio assets ready to showcase on GitHub and LinkedIn.